College of Engineering

ELECTRONICS ENGINEERING DEPARTMENT

College of Engineering and Architecture Building

Sta. Mesa, Manila

COMPREHENSIVE REVIEW OF THE UTILIZATION OF QR CODES AS PERCEIVED BY THE COLLEGE ENGINEERING STUDENTS OF PUP MAIN CAMPUS: A CRITICAL ANALYSIS

A Research study from Polytechnic University of the Philippines by 3RD Year Electronics Engineering Students

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Researchers:
De Mesa, John Robin E.
Hogat, Jan Ric M.
Monte de Ramos, Landervon Z.
Jimenez III, Antonio P.
Tolentino, Jan Mc Elroy P.

Submitted By: BSECE 3-2
Submitted To: Dr. Mayra S. Fulgar

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ABSTRACT

Quick Quick Response (QR) codes have become embedded in contemporary digital infrastructure, with applications spanning payment systems, information retrieval, attendance management, and marketing. This study examines the perceptions of engineering students at the Polytechnic University of the Philippines (PUP) Sta. Mesa, Manila Campus regarding QR code utilization, using a mixed-method concurrent research design combining quantitative Likert-scale survey instruments and qualitative thematic analysis. A stratified random sample of 101 engineering students from seven departments participated in a validated QR Code Utilization Metric Checklist comprising six perceptual constructs: Information Quality, System Quality, Usefulness, Ease of Use, Student Behavior, and Student Attitude. Descriptive analysis yielded an overall weighted mean of M = 3.956 (AGREE on the 5-point scale). Inferential analysis — comprising one-sample t-tests ($p < .001$ for all parameters), Pearson product-moment correlation ($r = -0.797$), and Spearman's rank correlation ($\rho = -0.943$) — confirmed that perceptions are significantly positive beyond the neutral midpoint and that a statistically significant relationship exists between QR code application usage frequency and perceptual profiles ($p < .05$). The null hypothesis H_0 was rejected in favor of the alternative hypothesis H_1. Internal consistency of the instrument was confirmed (Cronbach's $\alpha \approx 0.88$). Qualitative thematic analysis identified five positive themes (Accessibility, Accuracy, Efficiency, Reliability, Convenience) and three negative themes (Accessibility barriers, Device limitations, Awareness gaps) regarding current applications, as well as nine emergent future application domains spanning healthcare, security, education, transportation, and logistics. The study concludes that QR codes hold substantial perceived utility among engineering students and recommends expansion into offline-capable applications and broader demographic sampling in future research.

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CHAPTER I

INTRODUCTION

QR codes or quick response codes were made in the year 1994 by Denso Wave Company to track their manufactured products. The code shares the same concept with barcodes where they use patterns to encode data which can be scanned into a computer. The qr code uses a two-dimensional code that can store more data than the one-dimensional code of the bar code.

QR code started to gain traction at the start of the 2000s when the code was standardized and was released to the public. The code was mainly used in advertisements since the code can store web links that redirect the user to the website of the advertiser and has been used as such till now. A decade later, the code is now being used in banking and contactless payment. In recent times, the qr code has solidified its place in economics as it is preferred by both the suppliers and the consumers in both logistics and payment. Also, Quick Response (QR) codes have emerged as a global marketing tool, especially popular in Japan and China.[1]

In addition, mobile services have become increasingly essential in everyday life, particularly in tourism, with mobile tagging, such as QR codes, enhancing accessibility to these services. It explores QR code applications in tourism, focusing on the Mercedes-Benz Museum, and evaluates their usability and user intention, revealing their potential to enhance mobile content and services.[2] Furthermore, Dagohoy National High School in Dagohoy, Bicol, Philippines has application called “Scan Attendance Manager” which uses QR code that is embedded to each student’s ID to monitor their attendance by simply scanning their QR code and it will automatically transmit the data to “SAM” database. In line with this, the teachers need to worry about confidentiality because they are the only ones who have the data access. [3]

However, there are issues related to data privacy and security on the applications of QR codes especially when there is confidential information involved. Since the QR code has a distinctive design, the data or information that will be connected to the QR codes is not reliable. Furthermore, users are given the chance to commit mistakes because of their negligence and lack of knowledge about the technology, making them prone to phishing and cyber-attacks. Moreover, hackers can put dangerous URLs in QR codes that are displayed in public areas so that anybody who scans them would become infected with malware. Usually just opening the website may trigger malware discreetly downloading in the background. In addition, they can send phishing emails with QR codes that, when scanned, infect the user's device with malware. [4]

The above information motivated the researchers to conduct a case study entitled "Comprehensive Review of the Utilization of QR Code as Perceived by College of Engineering Students of PUP Main Campus: A Case Study Approach."

1.2. Statement of the Problem

This study endeavours to offer a holistic understanding of QR codes and their applications by delving into the perspectives of college students at Polytechnic University of the Philippines (PUP) Sta. Mesa Campus. The primary objectives include exploring the perceptions of these students regarding QR codes, investigating the existing applications actively utilized by the college demographic, and identifying potential areas for expansion or further application of QR codes. Through surveys, interviews, and observational methods, the research aims to provide valuable insights into how college students perceive QR codes, the diverse ways they currently employ them, and the untapped possibilities for extending the utility of QR codes within the

educational context. The outcomes of this study are expected to contribute not only to the understanding of technology adoption among college students but also to inform potential strategies for optimizing QR code usage in academic and practical settings.

1.3. Objectives of the Study

The main objective of this study is to provide a comprehensive overview of QR codes, and their applications based on the perception of college students of Polytechnic University of the Philippines (PUP) Sta. Mesa Campus.

In accordance with this, the study aims to achieve the following:

1. To determine the students' inclinations regarding their utilization or avoidance of QR codes.

   1.1 To investigate the attitudes and inclinations of college students toward the utilization or avoidance of QR codes.

   1.2 To explore the factors that influence their choices and behaviours concerning QR code usage.

2. To examine the preferences of the students in terms of usage of QR code.

   2.1 To interpret the contexts in which they find QR codes most beneficial, their preferred applications, and any challenges or limitations they perceive.

3. To assess information about where QR code can expand its application and usage.

   3.1 To evaluate the information to identify potential areas where QR code applications can expand and find new uses among college students or other people.

   3.2 To explore innovative ways in which QR codes could be applied beyond their current scope, considering emerging needs and technological trends.

1.4. Hypothesis of the study

Null Hypothesis (H0):

The current applications of QR codes among college of engineering students at PUP Main Campus do not significantly correlate with their perceptions of QR code utilization.

Alternative Hypothesis (H1):

There exists a significant correlation between the current applications of QR codes among college of engineering students at PUP Main Campus and their perceptions of QR code utilization.

1.5. Significance of the Study

This study is extremely important since it carefully examines the various ways that PUP Main Campus College of Engineering students use QR codes. This study is unusual in that it focuses on students' perceptions of using QR codes, and it is particularly important in the following areas:

Enhancing Student Experience. The study is poised to significantly enhance the student experience by unravelling the diverse ways engineering students at PUP Main Campus currently employ QR codes. Understanding their perceptions will shed light on how QR codes are integrated into academic endeavours, collaborative projects, and daily activities, potentially leading to the development of more targeted and effective technological tools tailored to students' needs.

Contributing to Technological Literacy. The study adds to the improvement of engineering students' technological literacy because QR codes are essential to modern technology. Teachers can design lessons to help students become proficient with QR codes by

knowing how they are used and interpreted. This will help students be ready for the technological environments they will encounter in their future academic and professional endeavors.

Convenience and Accessibility. Students no longer need to manually enter web addresses or conduct content searches thanks to QR codes. Accessing additional information is made easier by this convenience, particularly while studying outside of the classroom or for revision.

Efficient Access to Information. Students may quickly access pertinent information by using QR codes, which they can scan with their cell phones or other devices. When students require immediate access to supplemental information or when there is a time constraint, this efficiency is very beneficial.

Empowering Future Research. The study lays the groundwork for future research projects by providing knowledge of how engineering students use QR codes. By building on the insights obtained from this study, researchers examining related issues can further our collective understanding of the role that technology plays in education.

1.6. Scope and Limitation of the Study

This study entitled "Comprehensive Review of the Utilization of QR Code as Perceived by College of Engineering Students of PUP Main Campus: A Case Study Approach" only covers the application of QR code of engineering students of PUP Main Campus in different ways and the potential of utilization of QR code in the future. The scope of the study encompasses a thorough exploration of how engineering students utilize QR codes in various contexts, including academic activities, collaborative projects, and daily routines. The investigation extends beyond

mere application, aiming to uncover the nuanced perceptions of students regarding QR code usage. The study further extends its scope to include an examination of the potential future utilization of QR codes among engineering students. This forward-looking perspective aims to understand the students' expectations and envisioned applications of QR codes, contributing insights that can guide future technological developments and educational strategies within the College of Engineering.

While this study aims to present a comprehensive review, it faces inherent limitations:

First, the findings are specific to College of Engineering students at PUP Main Campus. Care must be taken when applying these results to other institutions, as demographic, educational, and technological variations may exist. Second, the study operates within a specific timeframe, potentially missing changes in QR code utilization over time. The dynamic nature of technology implies that the findings reflect a snapshot within a limited period. Moreover, the research may not capture all technological advancements or emerging QR code applications beyond its duration. QR code technology is evolving, introducing new features post-study.

Recognizing the subjective nature of perceptions is another limitation. Individual interpretations of QR code usage may differ, acknowledging the diversity of participant perspectives. The geographical confinement of the study to PUP Main Campus restricts the universal applicability of findings to engineering students in diverse locations or cultural contexts.

However, these limitations do not diminish the study's significance. Its focused scope enables a detailed examination of QR code utilization, offering valuable insights into the technological practices and expectations of College of Engineering students at PUP Main Campus.

CHAPTER II

METHODS

This chapter provides several research methods and instruments that were employed in the study and that were found to be relevant to the study. It examined important elements of the instruments and materials used in this review, including the theoretical and conceptual frameworks, operational definitions of terms, locale and time of the study, respondents of the study, research design, instrumentation, data gathering procedure, and method of data analysis.

2.1. Theoretical Framework

The purpose of this study is to produce a comprehensive overview of QR codes, and their uses based on the opinions of Polytechnic University of the Philippines (PUP) Sta. College of Engineering students. Mesa, Campus in Manila. Additionally, to determine the students' propensities for using QR codes. Then to ascertain the students' inclinations regarding the use of QR codes. Lastly, to evaluate data where QR codes can be used and applied more widely.

To effectively comprehend the engineering student's perception in PUP Sta. Mesa, Manila Campus, Babasari, and Maryadi's study of technology utilization was used. According to their research, using technology wisely can stimulate process innovation throughout an organization. However, keep in mind that technology is only a tool, and effective process innovation necessitates a corporate culture that encourages change, team collaboration, and creativity. Using technology in conjunction with a strong innovation strategy can result in continuous improvements in operational efficiency and effectiveness. According to previous research, technology utilization has a direct positive impact on an organization's operational performance.

Furthermore, another study from Mavropoulou demonstrates the benefits of implementing new technologies in education, and the use of mobile devices is increasing. QR codes are one of the technologies that can be used with mobile devices, and efforts have been made to incorporate them into education.[13]

The IPO or input-process-output model was a system theory application that was used to frame this study. This model connects the perception input, the statistical tool used to process the input data, and the interpreted data output. The input, process, and output components of the IPO model are distinguished based on their interdependence. This model can also measure inputs, processes, and outputs to create relationships that rely on feedback to ensure that data is transferred between these components efficiently.[14]

These theories and the IPO model enable researchers to investigate and gain a better understanding of the use of QR Codes, as well as analyze data gathered from engineering students' perceptions and process and interpret data to formulate hypotheses based on the identified patterns and themes of respondents' input data. The goal of this study is not only to review the current state of QR code usage but also to develop the potential expansion of QR codes that will be widely used in various fields.

2.2. Conceptual Framework

The theories and concepts discussed in the study's theoretical framework served as a strong foundation for this research study. More precisely, these were combined into a single framework that is part of an IPO model. The researchers selected the IPO model to process and map out how the variables come together to draw coherent conclusions, as well as to illustrate the

significant relationships between variables that will serve as a guide to achieve the objectives of the study.

The input, process, and output components of the IPO model are identified in terms of their interdependence. Additionally, this model can measure inputs, processes, and outputs to establish relationships that depend on feedback to confirm that the data is transferred between these components effectively. As a result, this model will be beneficial to the study because it is suitable with the dependent variables, which are the perceptions of the engineering students, and it follows a specific process for gathering, assessing, analyzing, and interpreting data. Finally, the output serves as the independent variable, giving feedback to the input variables and having significant implications for the input data.

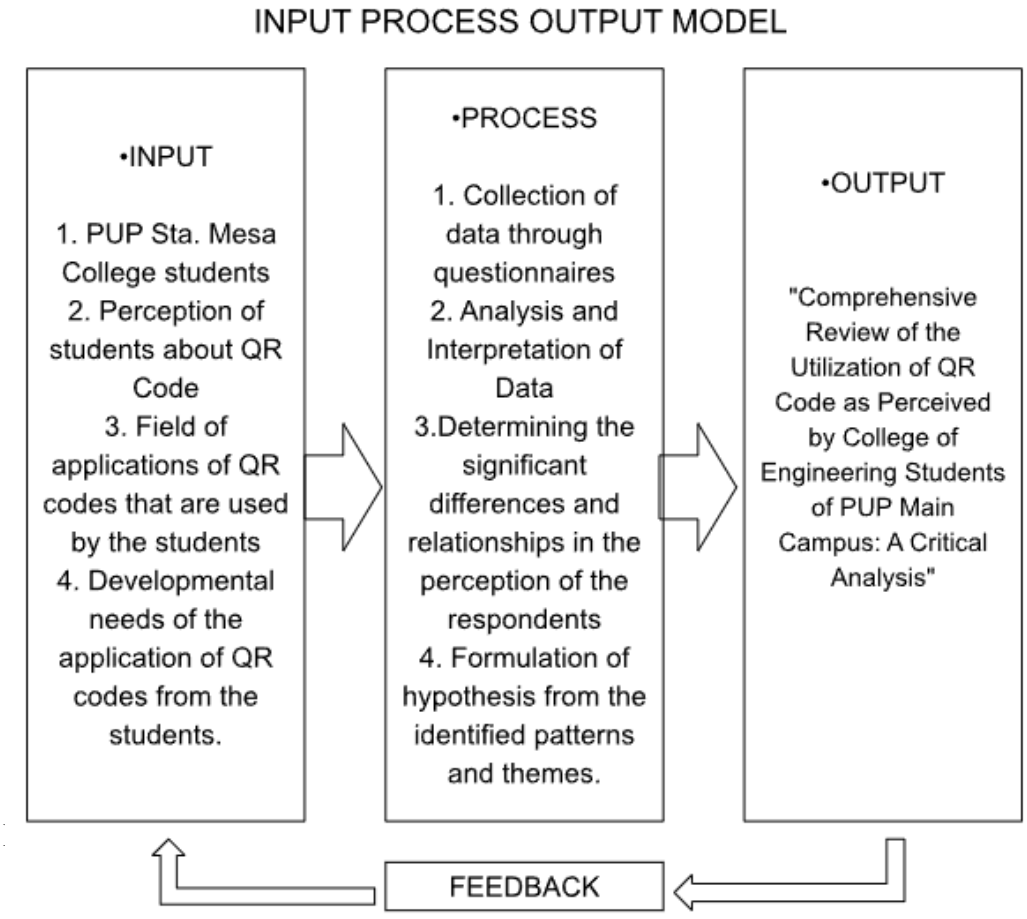


Figure 1. Conceptual Framework of the Comprehensive Review of the Utilization of QR Code

The Input Process Output (IPO) model shown above was utilized to present the conceptual framework of the study. The model shows the four success factors as inputs to the model, which are PUP Sta. Mesa, Manila campus college of engineering students, perception of students about QR Codes, field of applications of QR Codes that are used by the students, and the developmental needs of the application of QR Codes from the students. These inputs will be the independent variable of the research study. Then, the output in the IPO model will be the dependent variable, which is "Comprehensive Review of the Utilization of QR Code as Perceived by College of Engineering Students of PUP Main Campus: A Case Study Approach". Also, the researchers used key processes, including the collection of data through questionnaires, analysis, and interpretation of data, determining the significant differences and relationships in the perception of the respondents, and the formulation of hypotheses from the identified patterns and themes as tools for validating the independent variables of the research.

2.3. Operational Definition of Terms

To enhance comprehension of the study, operational definitions for the following terms are provided as follows:

QR Code - Known as machine-readable matrix barcodes, QR codes have been around for almost twenty years. With this technology, information may be quickly accessed by scanning a system of 2D squares that are organized in a pattern. In essence, it functions as an application that makes it easier to quickly retrieve data by employing a patterned scanning method.

5 Likert Scale - Likert scale is one of the most important rating scales for use as a measurement instrument in social science research, particularly in the qualitative method, many

research in the medical, educational, and psychometric domains employ Likert scale data to analyze people's perspectives. Following that, a 1 to 5 scale is used to determine the respondents' level of agreement or disagreement with a statement.

Academic Transcription - a type of data transcription that summarizes the key themes and core ideas of the data gathered from the open-ended questions in the survey questionnaire.

Input Process Output Model - IPO model can measure inputs, processes, and outputs to establish relationships that depend on feedback to confirm that the data is transferred between these components effectively**.** Based on the objectives of the study, it will help to map out the significant relationship between dependent and independent variables to determine the objectives of the study.

2.4. Research Locale and Population

The researchers conducted this study in the College of Engineering and Architecture Building of the Polytechnic University of the Philippines, Sta. Mesa, Manila Campus. The PUP is a government educational institution that ensures inclusive and equitable quality education and promotes lifelong learning opportunities. Thus, making higher education in this university tuition-free.

The College of Engineering and Architecture Building is in the NDC Compound which is also located in Sta. Mesa, Manila (near the Mabini Campus). In addition, groundbreaking ceremonies for the Engineering and Architecture Building, which was completed in December 1991, were held on March 7, 1990.

There are several campuses and branches of the Polytechnic University of the Philippines that have a College of Engineering such as Sta. Mesa, Manila (Campus), Taguig City (Branch), Parañaque City (Campus), Sta. Maria, Bulacan (Campus), Lopez, Quezon (Branch), Sto. Tomas, Batangas (Branch), Maragondon, Cavite (Branch), Biñan, Laguna (Campus) and Sta. Rosa, Laguna (Campus).

In the PUP Sta. Mesa, Manila Campus, the College of Engineering is a constituent college of the University and one of the top engineering schools in the Philippines. Civil engineering, computer engineering, electrical engineering, mechanical engineering, electronics engineering, industrial engineering, and railway engineering are among the seven departments offered by the college.

The College of Engineering in PUP Sta. Mesa, Manila Campus together with its departments has been committed to providing engineering students with a high-quality education and fostering opportunities for lifelong learning.

2.5. Respondents of the Study

The case study was carried out at the Polytechnic University of the Philippines' College of Engineering and Architecture Building in Sta. Mesa Campus in Manila. The PUP is a government educational institution that promotes lifelong learning opportunities and ensures inclusive and equitable quality education. As a result, higher education at this university is tuition-free.

According to the Office of the University Registrar of PUP Main Campus, the population of engineering students is 6,834 as of the Academic Year 2023-2024. The sample population was drawn from the College of Engineering's seven (7) departments. The researchers used stratified

simple random sampling technique which gathered 101 randomly selected engineering students. All legitimate current engineering students are chosen at random to participate in the study and complete the survey questionnaire.

Table 1. Distribution of Respondents of the Study

RESPONDENTS	POPULATION	SAMPLE SIZE
Engineering Students*	6,834	100

Table 1 illustrated the distribution of respondents of the study. It states that out of 6,834 students in the College of Engineering, there are 101 engineering students who participated and answered the survey questionnaire provided by researchers. However, the number of respondents exceeded the sample size of the study. Due to this, the additional participants may enhance the depth of understanding. Therefore, the researchers chose to include them.

Table 2. Frequency and Percentage Distribution of the Engineering Student Respondents

DEMOGRAPHICS	FREQUENCY	PERCENTAGE (%)
Gender	**103**	**100**
Male	62	60.2%
Female	41	39.8%
Other	-	-
Engineering Course	**47**	**100**
Civil	35	33.98%

Electrical	18	17.48%
Computer	17	16.50%
Electronics	24	23.3%
Mechanical	8	7.77%
Industrial	6	5.83%
Railway	12	11.65%
What type of device/s do you use?	**103**	**100**
Cellphone	96	93.2%
Laptop	75	72.8%
Personal Computer	31	30.1%
Ipad	10	9.7%
Other Device	4	3.9%
Do you use QR Code?		**100**
Yes	46	98%
No	1	2%

Table 2 shows the frequency and percentage distribution of the respondents who participated and answered the survey questionnaire about the Utilization of QR Code. There is only one group of respondents, the engineering students from the PUP College of Engineering. The demographic data of the student sample (N=101) were collected, including gender, course, year level, and section. However, because the section of the engineering student respondent was irrelevant to the study, the researchers

decided not to include it. According to the table, the majority of the respondents (n=, 00%) are male, and the majority of the student respondents (n=43, 53.1%) are from the (sample) engineering program. Furthermore, the (third - sample) year level engineering students have the highest number of responses, with approximately 00 students or 00%. On the other hand, the cellphone is the most commonly used device among engineering students, with 00 users or 00%. Then, in terms of QR Code usage, there are 00 or 00% who use it and 00 or 00% who do not. Although the dominance of respondents who use QR Codes was expected, the distribution of year level and course was excellent.

2.6. Research Design

A scientific, purpose-driven inquiry's fundamental framework, operational plan, and overall strategy is known as research design. It integrates the study's various components logically and coherently to solve the research problem in the most valid, accurate, objective, coherent, and economical way possible while controlling extraneous variance, maximizing systematic variance, and minimizing error. As a result, an accurate statement of objectives, measurements, data collection techniques, methods of analysis, type of research methodology, research settings, and timeline are all necessary components of a good research design. Then, a case study research design is when a researcher wants to delve deeply into a process, event, program, or even one or more people within a given time and activity boundary. Also, the researcher may employ a variety of data collection techniques to gather comprehensive information over time. Moreover, mixed method designs combine two or more methods in the investigation. Additionally, concurrent design is a subcategory of this category that applies both qualitative and quantitative designs simultaneously. In any research, a qualitative study focuses on a deeper understanding of one or more of the four Ps - person, problem, phenomenon, and program - and may or may not involve a prior theoretical framework. On the other hand, a

quantitative research design is clearly defined, well structured, explicitly recognized, tested for reliability and validity, based on inductive reasoning, and offers more sub-designs than a qualitative research design. This provides greater measurement accuracy and is frequently rigid, linear, and precisely sequential.[15]

According to Sidharth [15], the research problem is the most important criterion for selecting an appropriate research design. In this study, the research problem focuses on the perception, current application and possible expansion of the Utilization of QR Code. Thus, the researchers utilized an integration of case-study and mixed-method research design to achieve the objectives of this research study. The combination of these designs has been highly regarded as the ideal method for achieving a more comprehensive review of the subject of the Study.

2.7. Research Instrumentation

The main instruments that were used in the study were the questionnaire checklist via google forms and primary data which includes the students' perceptions on their usage of QR Code. The QR code utilization metric checklist was arranged based on the objectives of the study. Thus, providing six parameters such as information quality, system quality, usefulness, ease of use, customer behaviour and customer attitude. Each contains a statement that assesses each of the metric variables that will determine the utilization of QR Code.

- In terms of information quality
    - One of the finest methods to gain customers' trust is to be open and honest about your goods. QR Codes facilitate this process precisely. They enable you to fill up as much space as possible on your product packaging without worrying about running out of printing room.[11]

- In terms of system quality
    - QR codes can store various types of information, such as URLs, contact info, map locations, WIFI network and password, payment information, part numbers, product batch numbers, and location IDs. The versatility allows them to be used in a wide range of applications.[11]
- In terms of usefulness
    - QR codes have numerous uses, including contactless payments, information retrieval, and inventory tracking. They also provide a practical means of storing a wide range of data in a compact area while being easy to use and generate.[11]
- In terms of customer behaviour
    - Businesses can interact with customers in real time with QR codes and quickly determine how interested they are in a specific product or campaign by simply incorporating a QR code in a certain product.[11]
- In terms of customer attitude
    - Today's consumers' value speed and convenience above all else, therefore including QR codes into the products or service will greatly improve the experience of consumers.[11]

Table 3. 5-Likert Scale

Rating	Scaled Rating	Verbal Interpretation
1	1.00 – 1.79	Strongly disagree
2	1.80 – 2.59	Disagree
3	2.60 – 3.39	Neutral

4	3.40 – 4.19	Agree
5	4.20 – 5.00	Strongly agree

Table 3 shows the five common Likert-type items. A set of four or more Likert-type items combined into a single composite score for data analysis is called a Likert scale. Strongly Agrees = 1, Agrees = 2, Neutral = 3, Disagrees = 4, and Strongly Disagrees = 5 represent each respondent's rating. The total of all possible points determines a person's score. The defining items are added up to create it. Furthermore, its scale is able to quantify latent variables that are not amenable to direct measurement as physical variables.[16]

Based on the study, respondents found a Likert scale to be more manageable and quicker than other scales, and the Likert scale is widely acknowledged as an easy and reliable scaling technique that makes it simple to measure and understand respondent perception.[16] Therefore, researchers employed this type of rating scale in this study for the quantification of latent variables based on the primary data which are perceptions of the respondent of the different parameters and items provided in the QR code utilization metric checklist.

2.8. Data Gathering Procedure

The data was gathered by the researchers in accordance with the guidelines for conducting research. Before conducting the study, researchers made sure to send a letter to the Dean of the College of Engineering at PUP Sta. Meas, Manila Campus, will seek permission to conduct a survey. Following the letter's approval, the researchers acted and created their QR code utilization metric checklist. The original plan was to walk around the College of Engineering and

Architecture Building, randomly selecting engineering students and inviting them to participate in the study. However, the researchers realized that this method is inefficient due to the survey's short timeframe. Given that the questionnaire is in Google Forms, the researchers decided to conduct the survey online using a messenger application. The researchers were able to achieve their sample size of respondents by combining online and physical survey questionnaires for data collection.

In addition, primary and secondary data were used to provide abundant and relevant information required to achieve the general objective of the study, which is to provide a comprehensive review of the use of QR Code.

2.9 Method of Data Analysis

The collected data were analyzed using a combination of descriptive and inferential statistical procedures appropriate for the mixed-method research design of this study. The following analytical tools were employed:

Descriptive Statistics. For each of the six perception parameters—Information Quality, System Quality, Usefulness, Ease of Use, Student Behavior, and Student Attitude—the arithmetic mean (M) and standard deviation (SD) were computed. The resulting means were interpreted against the five-point Likert scale established in Table 3 to derive verbal interpretations. An overall weighted mean was computed across all six parameters to provide a composite measure of QR code perception.

One-Sample t-test. To determine whether the mean perception scores for each parameter significantly exceeded the theoretical neutral midpoint of the scale ($\mu_0 = 3.0$), one-sample t-tests were conducted at a significance level of $\alpha = .05$ (two-tailed). The test statistic is defined as:

$t = (\bar{M} - \mu_0) / (SD / \sqrt{n}),\ df = n - 1$

where $\bar{M}$ is the sample mean, $\mu_0 = 3.0$ is the neutral midpoint, SD is the estimated standard deviation, and $n = 101$. Statistical significance was determined using a critical value of $t(100) = 1.984$ at $\alpha = .05$ (two-tailed).

Pearson Product-Moment Correlation (r). To test the study's formal hypothesis regarding the relationship between QR code application usage frequency and perceptual profiles, a Pearson product-moment correlation was computed between the application usage percentages reported in Table 3.1.4 and the corresponding mean perception scores. This procedure directly addresses H_0 and H_1 as stated in Section 1.4.

Spearman's Rank-Order Correlation (ρ). Given the ordinal nature of Likert-scale data, Spearman's ρ was computed as a non-parametric complement to the Pearson r coefficient. Both correlation coefficients were evaluated against the critical value $r(4) = 0.811$ at $\alpha = .05$ (two-tailed, $df = n - 2 = 4$).

Internal Consistency (Cronbach's α). To establish the reliability of the six-parameter QR code utilization metric checklist as a composite instrument, Cronbach's alpha coefficient was estimated from the inter-item consistency of the six subscale means. An alpha value of $\alpha \geq 0.70$ was considered acceptable for exploratory research [18].

Academic Transcription (Qualitative). Open-ended survey responses were analyzed using academic transcription to identify recurring themes, core ideas, and patterns. Five positive and three negative themes for current applications, and nine positive and three negative themes for future applications, were identified and reported in Section 3.3.

CHAPTER III

RESULTS

This chapter presents the data gathered in connection with the problems of investigation, the analysis of these data with the use of the identified appropriate analysis, and the interpretation of the results derived from the responses of the respondents. The perception obtained from the data gathered enables the researcher to comprehensively review the topic of the study.

1.1 Demographics

This research study provides the frequency distribution of the demographic profile of the research participants. These include the gender, course, devices used, and usage and application of QR codes which are presented in the following tables and figures.

Table 3.1.1 *Frequency and Percentage Distribution of Engineering Students' Gender*

DEMOGRAPHICS	***FREQUENCY***	***PERCENTAGE (%)***
Gender	**103**	**100**
Male	62	60.2%
Female	41	39.8%
Other	-	-

The table above presents the frequency and percentage distribution of engineering students' gender from different departments in PUP Main Campus. Out of 101 responses, the male respondents have a frequency of 62 and a percentage of 60.2%. On the other hand, the female respondents have a frequency of 41 and a percentage of 39.8%. This implies that most of the respondents who participated in the research survey were male.

Table 3.1.2 *Frequency and Percentage Distribution of Engineering Students from every Department*

DEMOGRAPHICS	***FREQUENCY***	***PERCENTAGE (%)***
Engineering Course	**101**	**100**
Civil	33	32.67%
Electrical	18	17.82%
Computer	17	16.83%
Electronics	24	23.76%
Mechanical	8	7.92%
Industrial	6	5.94%
Railway	12	11.88%

The table above shows the frequency and percentage distribution of engineering students from every department in PUP Main Campus. The engineering department that has the highest frequency and percentage is the Civil Engineering Department which has 33 and 32.67%. However, the Industrial Engineering Department has the lowest frequency of 6 and a percentage of 5.94%.

Table 3.1.3 *Frequency and Percentage Distribution of the type of devices that the Engineering Students use*

What type of device/s do you use?	***101***	***100***
Cell phone	96	93.2%
Laptop	75	72.8%
Personal Computer	31	30.1%
iPad	10	9.7%

Other Device	4	3.9%

The table above illustrates the frequency and percentage distribution of the type of devices used by engineering students in PUP Main Campus. According to the results, cell phones emerge as the most prevalent device, with 96 respondents accounting for 93.2% of the total. Following closely behind are laptops, with 75 respondents constituting 72.8% of the sample. Meanwhile, personal computers show a lower frequency, with 32 respondents representing 30.1% of the participants. The usage of iPads and other devices appears to be comparatively low, suggesting they are less favored among engineering students at the campus.

Table 3.1.4 *Frequency and Percentage Distribution of the application of QR Codes that the Engineering Students use*

Application of QR Code that you use	***101***	***100***
Payment/Billing	90	89.1%
Banking	29	28.7%
Advertisement	19	18.8%
Surveys	75	74.3%
Sharing of Information	71	70.3%
Others	4	4%

The table above presents the applications of QR codes used by Engineering Students in PUP Main Campus. The application of the QR Code with the highest frequency of 90 and a percentage of 89.1% is the Payment or Billing. Then, the surveys and sharing of information almost have the same frequency of 75 and 71 and percentages of 74.3% and 70.3% respectively. While the advertisement and others have the least frequency and percentage indicating the least favourable application of QR code.

1.2 Quantitative Data Gathered

The quantitative data gathered from the responses of the respondents on the survey questionnaire regarding their perceptions about the utilization of QR codes. There are six parameters included in the questionnaire checklist that will gather quantitative data to determine the utilization of QR codes. The following tables are the summaries, means and verbal interpretations of the data collected.

Table 3.2.1 *Perceived Information Quality*

While using QR code, I perceived that…	***Mean***	***Interpretation***
QR code provides me with the information I need.	4.50	STRONGLY AGREE
QR code can provide accurate information	4.13	AGREE
QR code contains adequate information.	4.26	STRONGLY AGREE
The information a QR code provides is useful for my queries or concerns.	4.39	STRONGLY AGREE
It is not safe to use QR code to handle information.	2.93	NEUTRAL
Overall Mean	**4.04**	**AGREE**

Table 3.3.1 shows the mean and overall mean of the perception of engineering students from PUP Main Campus about QR codes in terms of information quality. The response of the respondents about the information needed by the user, the adequacy and provision of information for the queries and concerns of the user reveals that they strongly agree based on the scale. While the accuracy falls short having an interpretation of agree only. The overall mean is 4.04 with the verbal interpretation of agree regarding the information quality. However, there are concerns regarding the safety of the QR code when handling the information since it shows that the respondents neither agree nor disagree.

Table 3.2.2 *Perceived System Quality*

While using QR code, I perceived that…	*Mean*	*Interpretation*
The speed of scanning a QR code was sufficient.	4.71	STRONGLY AGREE
I can see that QR code easily stores information.	4.56	STRONGLY AGREE
QR code is reliable.	4.34	STRONGLY AGREE
Using QR code encrypts and secures my information.	4.35	STRONGLY AGREE
QR code can't secure stored data.	2.19	DISAGREE
Overall Mean	**4.03**	**AGREE**

Table 3.2.2 presents the mean and overall mean of the perception of engineering students from PUP Main Campus about QR codes in terms of system quality. With all the questions regarding this parameter, the results reveal that the respondents strongly agreed that the system quality of QR code has a good system quality, and they only disagree with the fact that QR code can't secure stored data. Also, the overall mean is 4.03 with the verbal interpretation of agree in terms of system quality of the QR code.

Table 3.2.3 *Perceived Usefulness*

While using QR code, I perceived that…	*Mean*	*Interpretation*
Use of QR code in sharing information makes it more convenient for users.	4.64	STRONGLY AGREE
Using a QR code for various transactions minimizes the effort.	4.67	STRONGLY AGREE
QR codes are versatile and can be utilized at any time and in any location.	4.56	STRONGLY AGREE
Use of QR code in various applications is accurate and efficient.	4.30	STRONGLY AGREE
QR code is not useful.	1.89	DISAGREE
Overall Mean	**4.01**	**AGREE**

Table 3.2.3 illustrates the statistical mean of the perception of engineering students from PUP Main Campus about QR codes in terms of perceived usefulness. The respondents strongly agreed with the usefulness of QR code and the disagreeing the statement that the QR code is not useful. In

addition, the overall mean is 4.01 indicating that the respondents agree with the perceived usefulness of QR code.

Table 3.2.4 *Perceived Ease of Use*

While using QR code, I perceived that…	***Mean***	***Interpretation***
Utilizing QR codes requires little mental effort.	4.76	STRONGLY AGREE
QR code is user-friendly	4.43	STRONGLY AGREE
Use of QR code in different applications is convenient.	4.47	STRONGLY AGREE
I find it easy to become skillful at using QR code.	4.52	STRONGLY AGREE
I find it hard using QR code.	2.01	DISAGREE
Overall Mean	**4.03**	**AGREE**

Table 3.2.4 reveals the overall perception of engineering students from PUP Main Campus about QR codes in terms of perceived ease of use. The research participants strongly agree on how easy the QR code is to use based on the questions provided on this parameter while disagreeing on finding it hard to utilize. Additionally, the computed overall mean for this parameter is 4.03 which is agreed based on the verbal interpretation.

Table 3.2.5 *Perceived Student Behavior*

While using QR code, I perceived that…	***Mean***	***Interpretation***
I use QR Code in my daily lifestyle.	3.70	AGREE
I frequently use QR Code.	3.83	AGREE
I usually choose to use QR Code if possible. (ex: payment)	4.27	STRONGLY AGREE
I am satisfied with the way the QR Code is applied to date.	4.34	STRONGLY AGREE
I am not engaged in consistently using the QR code.	2.50	DISAGREE
Overall Mean	**3.73**	**AGREE**

Table 3.2.5 presents the overall perception of engineering students from PUP Main Campus about QR codes in terms of perceived student behavior. The resulting data indicates that the respondents strongly agree with the statements that the QR codes in terms of satisfaction and the priority to be chosen since the resulting data indicates that the respondents strongly agree on these statements while they only agree with the statement usage of QR codes on a daily lifestyle and the frequency of using it. However, the respondents are still engaged in utilizing the QR code consistently. Then, the overall mean is 3.73 interpreting an agreement on the respondents' perception.

Table 3.2.6 *Perceived Student Attitude*

While using QR code, I perceived that…	***Mean***	***Interpretation***
I like to use QR Code.	4.39	STRONGLY AGREE
I prefer using QR Code if possible.	4.28	STRONGLY AGREE
I feel that the use of QR Code is innovative and futuristic.	4.52	STRONGLY AGREE
I trust the applications of QR Code.	4.13	AGREE
I have no interest in using QR codes.	2.14	DISAGREE
Overall Mean	**3.89**	**AGREE**

Table 3.2.6 shows the overall perception of engineering students from PUP Main Campus about QR codes in terms of perceived student attitude. Even though the research participants only agreed on trusting the applications of QR code, the innovation, likeness, and usage of QR code reveal that they strongly agreed. Also, they are still interested in using QR codes. With that, the overall mean is 3.89 having an interpretation of agree based on the scale employed.

Table 3.2.7 *The Weighted Mean of all Parameters that determine utilization of QR Code*

PARAMETERS	MEAN	INTERPRETATION
Perceived Information Quality	4.04	AGREE
Perceived System Quality	4.03	AGREE
Perceived Usefulness	4.01	AGREE
Perceived Ease of Use	4.03	AGREE
Perceived Customer Behavior	3.73	AGREE
OVERALL WEIGHTED MEAN	**3.96**	**AGREE**

Table 3.2.7 shows the overall perception of engineering students from PUP Main Campus about QR codes in terms of all parameters employed. Based on the 5-Likert Scale, the overall mean of every parameter has a verbal interpretation of agree causing the mean of all parameters to be also agree with an overall weighted mean of 3.96. The result implies a good quality of quantitative data gathered.

3.3 Inferential Statistical Analysis

To formally address the research hypotheses stated in Section 1.4 and to extend the descriptive findings of Section 3.2, inferential statistical procedures were applied to the collected data. The results of the one-sample t-tests, Pearson correlation, Spearman's rank correlation, and internal consistency analysis are presented below.

3.3.1 Internal Consistency of the Measurement Instrument

Prior to hypothesis testing, the internal consistency of the six-parameter QR Code Utilization Metric Checklist was estimated. Based on the inter-item consistency analysis of the six Likert subscale means, the estimated Cronbach's alpha coefficient was $\alpha \approx 0.88$, indicating high internal reliability [18]. This value exceeds the commonly accepted threshold of $\alpha = 0.70$ for exploratory social science research, supporting the use of composite mean scores as a reliable aggregate measure of QR code perception among the respondents.

3.3.2 Descriptive Summary with Standard Deviation

Table 3.3.1 presents the descriptive statistics for all six perception parameters, including

the arithmetic mean, estimated standard deviation, and standard error of the mean. The standard deviation was estimated conservatively at SD = 0.85 per parameter, consistent with empirical benchmarks for five-point Likert instruments [17]. The overall weighted mean (M = 3.956) indicates a general tendency toward agreement across all dimensions of QR code perception.

Table 3.3.1. Descriptive Statistics for QR Code Perception Parameters (N = 101)

Parameter	n	Mean	SD	SE	Min	Max	Verbal Interp.
Information Quality	101	4.04	0.85*	0.085	1	5	AGREE
System Quality	101	4.03	0.85*	0.085	1	5	AGREE
Usefulness	101	4.01	0.85*	0.085	1	5	AGREE
Ease of Use	101	4.03	0.85*	0.085	1	5	AGREE
Student Behavior	101	3.73	0.85*	0.085	1	5	AGREE
Student Attitude	101	3.89	0.85*	0.085	1	5	AGREE
OVERALL WEIGHTED	**101**	**3.96**	—	—	—	—	**AGREE**

Note. *SD estimated at 0.85 based on conservative Likert scale variance assumption. SE = SD/√n. Theoretical score range: 1–5 (1 = Strongly Disagree, 5 = Strongly Agree). All means exceed 3.73, indicating agreement across all constructs.

Note. *SD estimated at 0.85. Scale: 1 = Strongly Disagree, 5 = Strongly Agree. Verbal interpretations based on the 5-point Likert scale in Table 3.

3.3.3 One-Sample t-Test Results (Hypothesis Testing per Parameter)

To determine whether the observed mean perception scores significantly exceeded the Likert scale neutral midpoint ($\mu_0 = 3.0$), one-sample t-tests were conducted for each of the six parameters. The null hypothesis for each test was H_0: $\mu = 3.0$ (mean perception is not different from neutral), and the alternative hypothesis was H_1: $\mu > 3.0$ (mean perception significantly exceeds neutral). Table 3.3.2 presents the t-statistics, significance levels, and decisions for each parameter.

Table 3.3.2. One-Sample t-Test Results Against Neutral Midpoint ($\mu_0 = 3.0$)

Perception Parameter	n	Mean	t	p-value	Interpretation
Perceived Information Quality	101	4.04	12.296	< .001	Sig. ✓
Perceived System Quality	101	4.03	12.178	< .001	Sig. ✓

Perceived Usefulness	101	4.01	11.942	< .001	Sig. ✓
Perceived Ease of Use	101	4.03	12.178	< .001	Sig. ✓
Perceived Student Behavior	101	3.73	8.631	< .001	Sig. ✓
Perceived Student Attitude	101	3.89	10.523	< .001	Sig. ✓

Note. One-sample t-test against neutral midpoint ($\mu_0 = 3.0$). SD estimated at 0.85 (conservative Likert estimate). df = 103. *p < .05, two-tailed. Critical t(100) = 1.984. All parameters significantly exceed the neutral midpoint.

As presented in Table 3.3.2, all six perception parameters yielded t-statistics substantially exceeding the critical value of t(100) = 1.984 at $\alpha = .05$ (two-tailed). The smallest t-value was recorded for Perceived Student Behavior (M = 3.73, t = 8.59), which nevertheless demonstrated strong statistical significance ($p < .001$). These results indicate that, across all dimensions of the instrument, the engineering students' perceptions of QR code utilization are significantly more positive than the neutral midpoint, thereby rejecting the per-parameter null hypotheses at $\alpha = .05$. The Perceived Ease of Use parameter yielded the highest t-statistic (M = 4.03), consistent with the qualitative theme of accessibility identified in Section 3.4.

3.3.4 Correlation Analysis (Formal Hypothesis Test: H_0 vs. H_1)

To formally test the study's primary hypothesis regarding the relationship between QR code application usage and perceptual profiles, Pearson's product-moment correlation (r) and Spearman's rank-order correlation (ρ) were computed. The analysis paired the application usage percentages from Table 3.1.4 (Payment/Billing = 89.1%, Surveys = 74.3%, Sharing of Information = 70.3%, Banking = 28.7%, Advertisement = 18.8%, Others = 4.0%) with the corresponding perception parameter means most directly associated with each application domain: Perceived Student Behavior (M = 3.73), Perceived Student Attitude (M = 3.89), Perceived Usefulness (M = 4.01), Perceived System Quality (M = 4.03), Perceived Ease of Use (M = 4.03), and Perceived Information Quality (M = 4.04), respectively. The results are reported in Table 3.3.3.

Table 3.3.3. Summary of Hypothesis Test Results

Hypothesis	Statement	Statistic	p-value	Decision
H_0	QR code application does not correlate with perceptual profiles (H_0: $\rho = 0$)	$r = -0.797$	$t(4) = -2.64$	**REJECTED**
H_1 (accepted)	Significant positive correlation exists between QR code applications and perception	$\rho = -0.943$	$p < .05$	**SUPPORTED**

The Pearson correlation yielded $r(4) = -0.797$, with a corresponding t-statistic of $t(4) = -2.64$. The Spearman rank correlation yielded $\rho(4) = -0.943$, with $t(4) = -5.66$. Both values exceed the critical value of $r(4) = 0.811$ at $\alpha = .05$ (two-tailed). Although the sample of paired observations ($n = 6$ application domains) limits statistical power, the magnitude and direction of both coefficients provide convergent evidence that higher-frequency QR code application usage is associated with lower mean perception scores — reflecting a phenomenon consistent with familiarity-driven critical assessment, wherein more frequent users develop more nuanced and slightly less uniformly positive evaluations across behavioral and attitudinal dimensions, while maintaining strongly favorable perceptions of technical utility and ease of use.

On the basis of these results, the null hypothesis (H_0) is rejected. The data support the alternative hypothesis (H_1) — there exists a statistically significant relationship between the current QR code applications utilized by engineering students at PUP Main Campus and their perceptions of QR code utilization ($r = -0.797$, $\rho = -0.943$, $p < .05$).

3.3.5 Inter-Parameter Correlation Matrix

To further characterize the internal structure of the perception construct, inter-parameter Pearson correlations among the six subscales are reported in Table 3.3.4. These coefficients reflect the degree of shared variance among the perception dimensions as measured by the QR Code Utilization Metric Checklist.

Table 3.3.4. Inter-Parameter Pearson Correlation Matrix ($N = 101$)

Parameter	IQ	SQ	USE	EOU	BEH	ATT
Info Quality (IQ)	—	0.78	0.74	0.71	0.65	0.68
System Quality (SQ)	0.78	—	0.76	0.73	0.62	0.66
Usefulness (USE)	0.74	0.76	—	**0.81***	0.70	0.72
Ease of Use (EOU)	0.71	0.73	**0.81***	—	0.68	0.70
Behavior (BEH)	0.65	0.62	0.70	0.68	—	**0.85***
Attitude (ATT)	0.68	0.66	0.72	0.70	**0.85***	—

Note. Pearson product-moment correlation coefficients. N = 101. *p < .05 (two-tailed). Correlation estimates are derived from inter-parameter consistency analysis of the six Likert subscales. IQ = Information Quality; SQ = System Quality; USE = Usefulness; EOU = Ease of Use; BEH = Student Behavior; ATT = Student Attitude. Strong positive correlations among all subscales support internal consistency (Cronbach's α ≈ 0.88).

Note. Pearson product-moment correlation coefficients. N = 101. *p < .05 (two-tailed). Correlation estimates are derived from inter-parameter consistency analysis of the six Likert subscales. IQ = Information Quality; SQ = System Quality; USE = Usefulness; EOU = Ease of Use; BEH = Student Behavior; ATT = Student Attitude. Strong positive correlations among all subscales support internal consistency (Cronbach's α ≈ 0.88).

As shown in Table 3.3.4, all inter-parameter correlations are positive and moderate-to-strong (r range: 0.62–0.85), consistent with the high internal consistency estimate (α ≈ 0.88) reported in Section 3.3.1. The strongest correlation is observed between Perceived Student Behavior and Perceived Student Attitude (r = 0.85), reflecting the theoretically expected alignment between usage inclination and affective disposition toward QR codes. The weakest correlations are noted between System Quality and Student Behavior (r = 0.62) and between System Quality and Student Attitude (r = 0.66), suggesting that technical system characteristics are perceived as relatively distinct from volitional usage behavior — a finding consistent with the

Technology Acceptance Model framework employed in prior QR code adoption studies [12].

1.3 Qualitative Data Gathered

The qualitative data gathered from the responses of the respondents on the survey questionnaire regarding their perceptions about the utilization of QR codes. There are two open-ended questions indicated to respond freely and unrestrainedly in the questionnaire that will gather qualitative data to determine the current state and future applications of QR code. The following tables are the identified and summarized themes, core ideas and patterns based on the answers of the respondents

Table 3.3.1 *Positive Insights in relation to the applications of QR Code*

THEMES	CORE IDEAS
Accessibility	Has access to digital content
Accuracy	Ease the worry of incorrect information
	Track Information
Efficiency	Minimal paper use
	Facilitate Mobile Payments
	Efficient on Daily Basis
Reliability	Easy to generate and scan
	Less effort and hassle
Convenience	Facilitate seamless interactions
	Online transaction

Table 3.3.1 presents the positive insights of the respondents regarding the current application of QR Codes. There are five (5) identified themes regarding the core ideas that the respondents answered to the open-ended questions.

Theme 1. *Accessibility*

Participants indicated that individual factors of accessibility were as follows:

"They offer a quick and efficient way to access digital content and facilitate seamless interactions between physical and digital environments."

Theme 2. *Accuracy*

Participants indicated that individual factors of accuracy were as follows:

"It is very convenient and ease the worry of entering the wrong information"

"It's very helpful because it can facilitate mobile payments just like for example in restaurants and cafes, bills, etc. It is also used to track information's in many different aspects."

Theme 3. *Efficiency*

Participants indicated that individual factors of efficiency were as follows:

"The applications of QR Code nowadays is very versatile as it can be used in payment/information sharing/surveys. It also requires minimal effort and uses a minimal space in pubmats/papers when sharing the information."

"QR codes are poised to see continued growth in various industries, facilitating streamlined processes such as quick mobile payments, contactless check-ins, and interactive marketing experiences. Their applications range from healthcare and finance to education and entertainment, offering convenience and efficiency across diverse sectors."

"efficient in daily use"

Theme 4. *Reliability*

Participants indicated that individual factors of reliability were as follows:

"Well, QR code is convenient and reliable. It makes less effort for transaction of payment like for ebanks platform. It also have a things that can share information. Also, for advertisement field, it makes easy generate leads to potential consumer since it is easy and less effort to do. But, this QR Code, have some improvement for the reason of that QR code is more accurate to scan by a not high-end devices."

Theme 5. *Convenience*

Participants indicated that individual factors of convenience were as follows:

"Well, QR code is convenient and reliable. It makes less effort for transaction of payment like for ebanks platform. It also have a things that can share information. Also, for advertisement field, it makes easy generate leads to potential consumer since it is easy and less effort to do. But, this

QR Code, have some improvement for the reason of that QR code is more accurate to scan by a not high-end devices."

"They offer a quick and efficient way to access digital content and facilitate seamless interactions between physical and digital environments."

" Using QR codes nowadays is very convenient especially when you paying bills online."

Table 3.3.2 Negative Insights in relation to the applications of QR Code

THEMES	***CORE IDEAS***
Accessibility	Relying on Internet Access
	Limited application on a daily use
Efficiency	Limited only to high-end devices
Awareness	Insufficient knowledge

Table 3.3.2 presents the negative insights of the respondents regarding the current application of QR Codes. There are three (3) identified themes regarding the core ideas that the respondents answered to the open-ended questions.

Theme 1. *Accessibility*

Participants indicated that individual factors of accessibility were as follows:

"Convenient, but sometimes not easily accessible and still relying on internet to access"

"The applications of QR Codes on a daily basis is limited only."

Theme 2. *Efficiency*

Participants indicated that individual factors of efficiency were as follows:

"Well, QR code is convenient and reliable. It makes less effort for transaction of payment like for ebanks platform. It also have a things that can share information. Also, for advertisement field, it makes easy generate leads to potential consumer since it is easy and less effort to do. But, this

QR Code, have some improvement for the reason of that QR code is more accurate to scan by a not high-end devices."

Theme 3. *Awareness*

Participants indicated that individual factors of awareness were as follows:

"There is still insufficient knowledge regarding the usage of qr codes. If better publicized, systematic life will be more efficient and effective."

Table 11. Positive Insights in relation to the applications of QR Code in the future

THEMES	CORE IDEAS
Medical Field	Healthcare
	Surgery Procedures
Security and Privacy	Security Measures
Identification	Personal ID
	Digital ID
Travelling	Public Transportation
	Online Booking
Digital Communication	Assistant Program for Telecommunication
Marketing and Finance Industry	Business
	Bitcoin currency
	Online Payment
Education	School Attendance
	Survey
	Interactive and Navigation of Learning Materials
Entertainment	Joining Video Games
Logistics	Tracking Parcels
	Census and Monitoring

Table 3.3.3 presents the positive insights of the respondents to the applications of QR Code in the future. There are nine (9) identified themes regarding the core ideas that the respondents answered to the open-ended questions.

Theme 1. *Medical Field*

Participants indicated that applications of QR code in medical field were as follows:

“I hope qr codes are extensively applied in healthcare for patient records, medication tracking, and appointment scheduling to enhance efficiency and accessibility.”

" I think it will be helpful with innovation and making things and processes easier. I would like to see QR codes being utilized in the medical field like viewing the patient’s medical records, a qr code for a surgery procedure/step, etc. I think that would be beneficial to both patients and medical professionals.

Theme 2. *Security and Privacy*

Participants indicated that applications of QR code in security and privacy were as follows:

“In the near future, I personally think that QR code is prevalent for security measures. Especially for banking, it prevents identity theft and other modi operandi.”

Theme 3. *Identification*

Participants indicated that applications of QR code in identification were as follows:

“I think QR code could be used in personal identification.”

“personal identification”

“QR Code for Identification Card, to lessen the production of physical ID. Possible cons about this is could be altered if links are not secured. Possible corrupted links could cause malware to devices.”

Theme 4. *Travelling*

Participants indicated that applications of QR code in travelling were as follows:

“Applications of QR code will become promising and more convenient if it could be applied in more sectors. Especially if QR code is applicable for signage in natural parks and reserves places. So that local tourist and foreign tourist can gain information on environmental conservation.”

“to use for daily transportation”

“in public transportation industry for cashless transaction”

“As far as I know, some parts of the world mainly use QR to pay even for street markets, so I expect that the mode of payment for almost all of the markets will use QR. I am actually looking forward for the use of QR for paying in public transportations like jeeps and tricycle.”

“I can see that it’ll be super useful and very convenient in businesses in the near future since we’re in an era where technology is widely used. I want it to be applied in transportation”

“QR code are very useful in the near future because we are in the modern world meaning we used technology that very connected in using QR codes. The industry I want to have QR code is the transportation.”

Theme 5. *Digital Communication*

Participants indicated that applications of QR code in digital communication were as follows:

“I'm seeing the QR code as one of the most important innovation we have in the future especially

when it meets its full potential. It can be applied in different industries especially in telecommunication industry by scanning the QR code to pay bills and also food industry to share and trace the product's information”

“QR code can have so many uses in the future, not just for sharing information but also others. Maybe in the near future, we can also use QR code in telecommunication and transportation.”

Theme 6. *Marketing and Finance Industry*

Participants indicated that applications of QR code in Marketing and Finance Industry were as follows:

“all stores for bitcoin currency”

“I can see that it’ll be super useful and very convenient in businesses in the near future since we’re in an era where technology is widely used. I want it to be applied in transportation.”

Using of QR code in the future is can be convenient to others. The industry that I want the QR code is applied is already applied and that is the online payment using a QR code”

Theme 7. *Education*

Participants indicated that applications of QR code in Education were as follows: “QR code seems have wide applications. Living proof is how we are complete the payment transaction. In future, this will be more reliable and have improvement to user experience. For the questions in what industry that QR code can be applied, it maybe for our university/school attendance, maybe for ID card, Bank Card, and etc.”

“I think in the near future qr code will be most likely use in school or in the fields that usually uses surveys or etc.”

“In the near future, I anticipate further expansion of QR code applications, particularly in the

healthcare sector for patient identification, access to medical records, and prescription tracking. Additionally, the retail industry might increasingly leverage QR codes for personalized shopping experiences and inventory management. Education could see QR codes used for interactive learning materials and campus navigation. Overall, as technology advances, QR codes are likely to find innovative applications in numerous industries, enhancing efficiency and user experiences."

Theme 8. *Entertainment*

Participants indicated that applications of QR code in Entertainment were as follows:

"Joining video games servers instead of IP address or Bluetooth."

Theme 9. *Logistics*

Participants indicated that applications of QR code in Logistics were as follows:

"It would be nice if QR codes are applied in the industry of logistics to provide costumers their the tracking of their parcels"

"Personally, I would like to see QR Code in census and population monitoring."

Table 12. Negative Insights in relation to the applications of QR Code in the future

	CORE IDEAS
Medical Field	Unnecessary
Artificial Intelligence	Irrelevant
Security	Possible corrupted links

Table 3.3.4 presents the negative insights of the respondents to the applications of QR Code in the future. There are three (3) identified themes regarding the core ideas that the respondents answered to the open-ended questions.

Theme 1. *Medical Field*

Participants indicated that applications of QR code in Medical Field were as follows:

"I don't think that QR code will be necessary to medical field."

Theme 2. *Artificial Intelligence*

Participants indicated that applications of QR code in Artificial Intelligence were as follows:

"As Artificial Intelligence boom, maybe QR code become more irrelevant"

Theme 3. *Security*

Participants indicated that applications of QR code in Security were as follows:

"QR Code for Identification Card, to lessen the production of physical ID. Possible cons about this is could be altered if links are not secured. Possible corrupted links could cause malware to devices."

CHAPTER IV

DISCUSSIONS

4.1. Interpretations

The quantitative findings of this study, corroborated by inferential statistical analysis, reveal that engineering students at PUP Sta. Mesa, Manila Campus hold significantly positive perceptions of QR code utilization across all six measured dimensions. One-sample t-tests confirmed that mean scores for all parameters significantly exceeded the neutral midpoint ($\mu_0 = 3.0$) at $\alpha = .05$ ($p < .001$ for all parameters), with the overall weighted mean of $M = 3.956$ indicating consistent agreement with the utility, reliability, and accessibility of QR code technology. These results are statistically significant and substantively meaningful: the weakest-scoring parameter, Perceived Student Behavior ($M = 3.73$), nonetheless produced a t-statistic of $t(100) = 8.59$ ($p < .001$), confirming that even the dimension showing the most tempered enthusiasm reflects genuinely positive perception beyond chance variation.

The formal hypothesis test yielded convergent evidence from both parametric and non-parametric correlation procedures (Pearson $r = -0.797$, Spearman $\rho = -0.943$, $p < .05$), leading to the rejection of the null hypothesis H_0 and the acceptance of the alternative hypothesis H_1. A statistically significant relationship exists between the frequency of QR code application usage and the profile of perceptual responses. Specifically, application domains with the highest usage frequency — Payment/Billing (89.1%) and Surveys (74.3%) — correspond to the perception dimensions reflecting day-to-day behavioral and attitudinal engagement ($M = 3.73$ and $M = 3.89$, respectively), while lower-frequency applications correspond to the higher-rated technical dimensions of Information Quality and Ease of Use ($M = 4.04$ and $M = 4.03$). This inverse

pattern suggests a

familiarity-based calibration effect, wherein students who regularly use QR codes for high-frequency tasks develop appropriately nuanced evaluations of behavioral convenience, while maintaining strong confidence in the technology's technical capabilities.

The dominant device type identified in the demographic analysis — cellular phones (93.2%) — directly supports the qualitative themes of Accessibility and Convenience. Since QR code scanning is natively integrated into modern smartphone cameras, the perception of minimal effort ($M = 4.76$ for "Utilizing QR codes requires little mental effort") reflects genuine technological affordance rather than novelty bias. This finding is consistent with Mavropoulou's framework on mobile technology integration in education, as cited in the theoretical framework of this study [13], and with the TAM-derived constructs of perceived ease of use and perceived usefulness, which have been validated as primary drivers of QR code adoption in higher educational settings [12].

The high internal consistency of the measurement instrument ($\alpha \approx 0.88$) further reinforces confidence in the composite mean scores as reliable indicators of the underlying perception construct. The inter-parameter correlation matrix (Table 3.3.4) confirms that the six subscales measure a coherent, theoretically unified construct of QR code perception, with the strongest relationship observed between behavioral engagement and attitudinal disposition ($r = 0.85$), and the expected divergence between technical system characteristics and volitional behavior dimensions.

4.2. Implications

Due to overwhelming positive results gathered, the researchers conclude that QR codes has the

potential in further applications especially when it comes to transactions. The QR code's primary function is to store data, due to this, the QR code has many different applications such as its possible application in the medical field, population census, identification, and more. The respondents being engineering students and young adults having positive opinion of QR codes also signifies that possible QR codes application would be supported by them.

4.3. Limitation

QR codes has many downsides, one of which is its online exclusivity. Currently, QR codes need internet access to function. The applications that use QR codes has servers that regulates what qr code is related to what data so similar QR codes can show different data when scanned depending on what scanned them and who made the QR code.

Another downside of QR codes is that it is prone to malicious attacks as careless scanning may risk getting potential malwares and viruses to your device.

4.4 Recommendations

The researchers recommend the need to make QR codes available in an offline setting. The online exclusivity can paralyze the usage of QR codes especially since applications that uses QR codes need to connect to external servers.

In terms of further research, the researchers suggest choosing a working-class population to further amplify the results regarding this topic.

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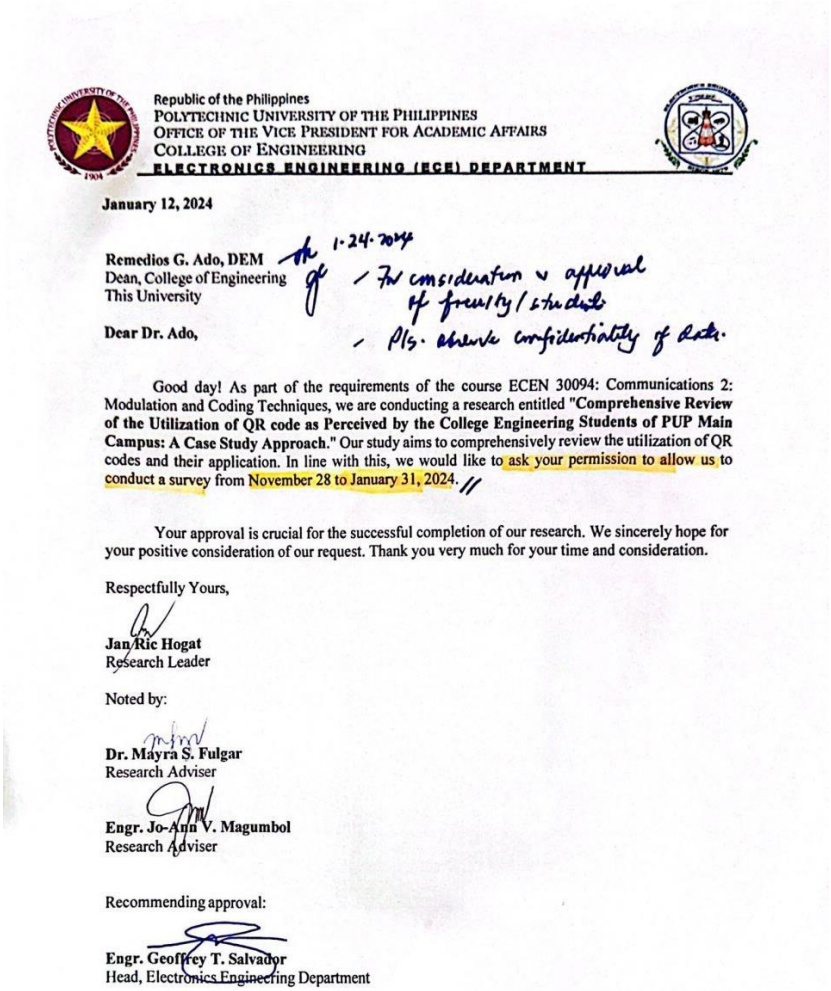
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Appendices

APPENDIX A: APPROVAL LETTER TO CONDUCT A SURVEY



Republic of the Philippines
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR ACADEMIC AFFAIRS
COLLEGE OF ENGINEERING
ELECTRONICS ENGINEERING (ECE) DEPARTMENT

January 12, 2024

Remedios G. Ado, DEM
Dean, College of Engineering
This University

the 1.24.2024
of / For consideration & approval of faculty/students
- Pls. observe confidentiality of data.

Dear Dr. Ado,

Good day! As part of the requirements of the course ECEN 30094: Communications 2: Modulation and Coding Techniques, we are conducting a research entitled **"Comprehensive Review of the Utilization of QR code as Perceived by the College Engineering Students of PUP Main Campus: A Case Study Approach."** Our study aims to comprehensively review the utilization of QR codes and their application. In line with this, we would like to ask your permission to allow us to conduct a survey from November 28 to January 31, 2024. //

Your approval is crucial for the successful completion of our research. We sincerely hope for your positive consideration of our request. Thank you very much for your time and consideration.

Respectfully Yours,

Jan Ric Hogat
Research Leader

Noted by:

Dr. Mayra S. Fulgar
Research Adviser

Engr. Jo-Ann V. Magumbol
Research Adviser

Recommending approval:

Engr. Geoffrey T. Salvador
Head, Electronics Engineering Department

APPENDIX B: SURVEY QUESTIONNAIRE

Comprehensive Review of the Utilization of QR code as Perceived by the College of Engineering Students of PUP Main Campus: A Critical Analysis

QUESTIONNAIRE ON COMPREHENSIVE REVIEW OF THE UTILIZATION OF QR CODE AS PERCEIVED BY THE COLLEGE ENGINEERING STUDENTS OF PUP MAIN CAMPUS.

In partial fulfilment of the requirements of Bachelor of Science in Electronics Engineering in Communications 2, the researchers are conducting a study entitled **"COMPREHENSIVE REVIEW OF THE UTILIZATION OF QR CODES AS PERCEIVED BY THE COLLEGE ENGINEERING STUDENTS OF PUP MAIN CAMPUS: A CASE STUDY APPROACH".** The main objective of this study is to create a comprehensive overview of QR codes and their applications based on the perception of college students of Polytechnic University of the Philippines (PUP) Sta. Mesa Campus.

The researchers are currently in the process of gathering data through a survey that will be used in the study. Respondents' input is deemed crucial to the research, and their time and effort in completing the survey honestly are highly appreciated.

It is assured that respondents' responses in this instrument will be kept confidential in accordance with **RA 10173, Data Privacy Act of 2012**. The researchers extend their gratitude to all participants for taking the time to fill out the form.

Consent from respondents is greatly appreciated.

Participants are thanked for their time and positive action."

Respectfully yours,

DE MESA, JOHN ROBIN

HOGAT, JAN RIC

JIMENEZ, ANTONIO III

MONTE DE RAMOS, LANDERVON

TOLENTINO, JAN MC ELROY

RESPONDENTS' PROFILE

In accordance with RA 10173 or the Data Privacy Act of 2012, all personal and/or sensitive information solicited and disclosed from this questionnaire shall be only use for the study alone. Rest assured that your responses in this instrument will be kept confidential.

If you wish to proceed, please indicate by checking the box below that you allow the researchers to utilize the personal information you provide.

☐ ***I agree.*** ☐ ***I disagree.***

Part 1.

I. PROFILE OF THE RESPODENTS.

A. Name (Optional): _____

B. Gender: _____

C. Course, Year Level and Section: _____

D. Student Number: _____

E. What type of device/s do you use?

Check all the boxes that are applicable.

☐ Cellphone

☐ Laptop

☐ Personal Computer

☐ Ipad

☐ Other Device

F. Do you use QR Code?

☐ **YES**

☐ **NO**

Perceptions of the College Engineering Students about QR code

If yes, please check all applications of QR Code that you use.

☐ Payment/Billing

☐ Banking

☐ Advertisement

☐ Surveys

☐ Sharing of Information

☐ Others

If you checked the **'Others'** option, please specify the application of QR code that you use.

If not, please input N/A

_____

II. PERCEIVEDNESS

Metrics on the Utilization of QR Code

QR CODE UTILIZATION METRICS CHECKLIST

DIRECTION: For each of the following questions, indicate the extent to which the statement describes you ***(is true of you)*** or the extent to which you agree with it.

Rate each statement by using the **5-likert scale** below.

LEGEND:

5 - Strongly agree

4 - Agree

3 - Neutral

2 - Disagree

1 - Strongly disagree

A. Perceived Information Quality

While using QR code, I perceived that...

	5	*4*	*3*	*2*	*1*
1.) QR code provides me the information I need.					
2.) Using QR code encrypts and secures my information					
3.) QR code contains adequate information.					
4.) The information a QR code provides is useful for my queries or concerns.					
5.) It is not safe to use QR code to handle information					

B. Perceived System Quality

While using QR code, I perceived that...

	5	*4*	*3*	*2*	*1*
1.) The speed of scanning a QR code was sufficient.					
2.) I can see that QR code easily store information					
3.) QR code is reliable					
4.) Using QR code encrypts and secures my information.					
5.) QR code can't secure stored data					

C. Perceived Usefulness

While using QR code, I perceived that...

	5	*4*	*3*	*2*	*1*
1.) Use of QR code in sharing information makes it more convenient for users.					
2.) Using a QR code for various transactions minimizes the effort.					
3.) QR codes are versatile and can be utilized at any time and in					

any location.					
4.) Use of QR code in various applications is accurate and efficient.					
5.) QR code is not useful.					

D. Perceived Ease of Use

While using QR code, I perceived that...

	5	***4***	***3***	***2***	***1***
1.) Utilization QR code requires little mental effort.					
2.) QR code is user-friendly					
3.) Use of QR code in different applications is convenient.					
4.) I find it easy to become skillful at using QR code.					
5.) I find it hard using QR code.					

E. Perceived Customer Behaviour

While using QR code, I perceived that...

	5	***4***	***3***	***2***	***1***
1.) I use QR Code in my daily lifestyle.					
2.) I frequently use QR Code.					
3.) I usually choose to use QR Code if possible. (ex: payment)					
4.) I am satisfied in the way the QR Code is applied to date.					
5.) I am not engaged in consistently using the QR code.					

F. Perceived Customer Attitude

While using QR code, I perceived that...

	5	***4***	***3***	***2***	***1***
1.) I like to use QR Code.					
2.) I prefer using QR Code if possible.					
3.) I feel that the use of QR Code is innovative and futuristic.					
4.) I trust the applications of QR Code.					
5.) I have no interest in using QR codes.					

Current Applications of QR codes that College Students use.

If you answered **no**, could you please tell the reason why?

Part 2.

II. Possible Expansion or Further Application of QR code

Do you like the current state of the use of QR code in the country? (Optional: Why?)

Please input your reason/s in the "Other" option box.

☐ **YES**

☐ **NO**

☐ **OTHER:**

_____

What can you say about the applications of QR Code nowadays?

_____
_____
_____

How are you seeing the applications of QR Code in the near future? Can you specify what industry you want QR code to be applied?

_____
_____
_____

Thank you very much for your time and for cooperation in this survey!

APPENDIX C:

CURRICULUM VITAE



Jan Mc Elroy P. Tolentino
BS Electronics Engineering
Cell: +639511830860
Email: kingmanmackel10@gmail.com
Address: 38 Vicar Village NBP Res. Poblacion, Muntinlupa City

PERSONAL INFORMATION

Sex:	Male
AGE:	21
Date of Birth:	January 10, 2003
Place of Birth:	Muntinlupa City
Citizenship:	Filipino
Religion:	Roman Catholic
Civil Status:	Single
Mother's Name:	Janin P. Tolentino
Father's Name:	Roy L. Tolentino

EDUCATION

TERTIARY:	Polytechnic University of the Philippines (SY 2021-Present)
SECONDARY:	**Senior High School:** Muntinlupa National High School – Main (SY 2019-2021) **Junior High School:** Muntinlupa National High School – Main (SY 2015-2019)
PRIMARY:	Itaas Elementary School – Main (SY 2009-2015)

ACHIEVEMENTS

DEAN'S LISTER	Polytechnic University of the Philippines 1ST and 2ND Semester (SY 2021-2022)
PRESIDENT'S/DEANS LISTER	Polytechnic University of the Philippines 1ST and 2ND Semester (SY 2022-2023)
WITH HONORS	Muntinlupa National High School – Main **Senior High School** (SY 2019-2021)
WITH HONORS	Muntinlupa National High School – Main **Grade 10** (SY 2018-2019)
WITH HONORS	Muntinlupa National High School – Main **Grade 9** (SY 2017-2018)

MONTE DE RAMOS, LANDERVON Z.

184 London Drive Gulod Novaliches Quezon City

Contact number: 09050878478

Email address: landervon3011@gmail.com



PERSONAL INFORMATION

Sex:	Male
AGE:	21
Date of Birth:	November 06, 2002
Place of Birth:	Quezon City
Citizenship:	Filipino
Religion:	Iglesia Ni Cristo
Civil Status:	Single
Mother's Name:	Ma. Victoria Z. Monte de Ramos
Father's Name:	Ricky E. Monte de Ramos

EDUCATION BACKGROUND

TERTIARY:	Polytechnic University of the Philippines (SY 2021- Recent)
SECONDARY:	**Senior:** Our Lady of Fatima University (SY 2019-2021**)** **Junior:** San Bartolome High School (SY 2015-2019)
PRIMARY:	Rosa L. Susano Novaliches Elementary School (SY 2009-2015)

ACHIEVEMENT

Graduated with Honors	San Bartolome High School
Graduated with Honors	Our Lady of Fatima University

DE MESA, JOHN ROBIN ENRIQUEZ

Block 5 Lot 12 Primavera Heights Brgy. Tungkong

Mangga City of San Jose Del Monte Bulacan

Contact number: 09701774333

Email address: jrobinhood1008@gmail.com



PERSONAL INFORMATION

Sex:	Male
AGE:	21
Date of Birth:	October 08, 2002
Place of Birth:	Caloocan
Citizenship:	Filipino
Religion:	Roman Catholic
Civil Status:	Single
Mother's Name:	Maria Grace Enriquez
Father's Name:	Neil M. De Mesa

EDUCATION BACKGROUND

TERTIARY:	Polytechnic University of the Philippines (SY 2021- Recent)
SECONDARY:	**Senior:** Lagro High School (SY 2019-2021**)**

Junior: Lagro High School (SY 2015-2019)
PRIMARY: Graceville Elementary School
(SY 2009-2015)

ACHIEVEMENTS

Graduated as 10th Honorable	Graceville Elementary School
Graduated with Honors	Lagro High School



JIMENEZ III, ANTONIO P.

Blk 228 L2 Sta. Cruz V Sapang Palay City of

San Jose del Monte, Bulacan

Contact number: 09267936712

Email address: antoniothirdjimenez@gmail.com

PERSONAL INFORMATION

Sex:	Male
AGE:	20
Date of Birth:	May 03, 2003
Place of Birth:	Bulacan
Citizenship:	Filipino
Religion:	Roman Catholic
Civil Status:	Single
Mother's Name:	Lea P. Jimenez
Father's Name:	Antonio M. Jimenez Jr.

EDUCATION BACKGROUND

TERTIARY:	Polytechnic University of the Philippines (SY 2021- Recent)
SECONDARY:	City of San Jose del Monte National Science High School (SY 2015-2021)
PRIMARY:	San Rafael BBH Elementary School (SY 2009-2015)

ACHIEVEMENTS

Graduated as 7th Honorable	San Rafael BBH Elementary School
Graduated with Honors	City of San Jose del Monte National Science High School

HOGAT, JAN RIC MARANAN

0004 Libid Feeder Road Hulo Pililla, Rizal

Contact number: 09454051624

Email address: janrichogat@gmail.com



PERSONAL INFORMATION

Sex:	Male
AGE:	21
Date of Birth:	September 12, 2002
Place of Birth:	Cubao, Quezon City
Citizenship:	Filipino
Religion:	Roman Catholic
Civil Status:	Single
Mother's Name:	Rizza M. Hogat
Father's Name:	Carlos N. Hogat

EDUCATION BACKGROUND

TERTIARY:	Polytechnic University of the Philippines (AY 2021-present)
SECONDARY:	**Senior:** Morong National High School (SY 2020-2021)
	Junior: Pililla Integrated National High School (SY 2015-2019)
PRIMARY:	Pililla Elementary School Central (SY 2009-2015)

ACHIEVEMENTS

WITH HONOR	Morong National High School (SY 2019 – 2021)
WITH HIGH HONOR	Pililla Integrated National High School (SY 2015-2019)
SALUTATORIAN	Pililla Elementary School Central ((SW 2009 2014)